

Unit 1, Lesson 4: Multiplying Multi-Digit Numbers With Decimals to the Thousandths



Benchmark

MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.



Learning Target

- ☐ I can fluently multiply multi-digit numbers with decimals to the thousandths.



1.4.1 Warm-Up: Decimal Products

- Place a digit in each blue box so that the product is as close to 50 as possible. Use only digits 1 to 9, at most one time each.

$$\boxed{}.\boxed{} \times \boxed{}.\boxed{} =$$



1.4.2 Refresher: Reasoning with Powers of Ten

Several expressions are shown.

$250 \cdot 0.01$	$250 \div 100$	$250 \cdot 0.001$	$250 \cdot \frac{1}{100}$	$250 \cdot 0.1$
$250 \div 1000$	$250 \cdot \frac{1}{10}$	$250 \cdot \frac{1}{1000}$	$250 \div 10$	

- Work with your partner to complete the table by writing the expressions that match each value indicated.

Value	Equivalent Expressions		
25			
2.5			
0.25			

- When working together through the problem, Cheryl and Miranda had a disagreement.

Cheryl said that $\frac{1}{100}$ is equivalent to 0.001 because both have two zeros.	Miranda said that $\frac{1}{100}$ is equivalent to 0.01 because of place value.
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- Which student do you agree with?
- Explain why you agree.

Recall that fractions like $\frac{1}{10}$, $\frac{1}{100}$, and $\frac{1}{1000}$ can be used to find the location of the decimal point in terms of place value when multiplying decimals.

For example, 24×0.02 can be thought of as $24 \times \frac{2}{100}$ or $24 \times 2 \times \frac{1}{100}$. This is equal to $\frac{48}{100}$ which is read, "48 hundredths" and written as 0.48 in decimal form with the 8 in the hundredths place.

$$24 \times 0.02 = 0.48$$

- With your partner, complete the equations and statements below to show how similar thinking can be used to find the products of decimals.

a.

$$(0.7) \cdot (0.09) = (7 \times \frac{1}{10}) \cdot (9 \times \frac{1}{100}) = (7 \cdot 9) \cdot (\frac{1}{10} \cdot \frac{1}{100}) = \boxed{} \cdot \frac{1}{\boxed{}}$$

Written as a decimal, $(0.7) \cdot (0.09) = \underline{\hspace{2cm}}$.

b.

$$(1.5) \cdot (0.43) = (15 \times \frac{1}{10}) \cdot (43 \times \frac{1}{100}) = \boxed{} \cdot \frac{1}{\boxed{}}$$

Written as a decimal, $(1.5) \cdot (0.43) = \underline{\hspace{2cm}}$.

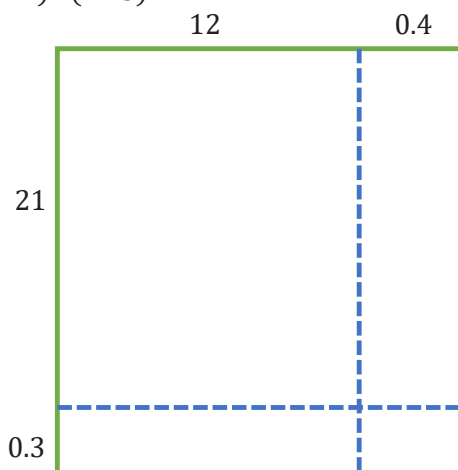




1.4.3 Guided Instruction: Multiplying Decimals to the Thousandths

Different strategies can be used to find the product of decimals to the thousandths.

- Byanca prefers using an area model to find the product of decimals. Below is an area model that Byanca drew to represent $(12.4) \cdot (21.3)$.



- Find the region that represents $(0.4) \cdot (0.3)$. Shade that region in the diagram and label it with its area of 0.12.
- Label the other regions of the model with their respective areas.
- Find the value of $(12.4) \cdot (21.3)$ using the labeled area model. Show your work.

- Sergio prefers to use partial products to find the value of $(12.4) \cdot (21.3)$, while Alicia prefers using the standard algorithm. Their work is shown.

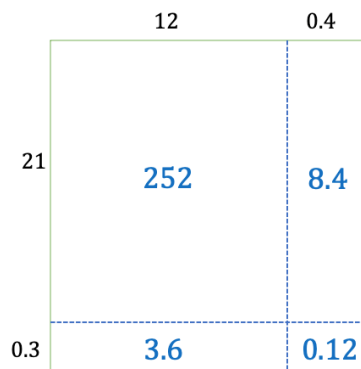
Sergio's Work Partial Products	Alicia's Work Standard Algorithm
$ \begin{array}{r} 12.4 \\ \times 21.3 \\ \hline 0.12 \\ 0.6 \\ 3 \\ 0.4 \\ 2 \\ 10 \\ 8 \\ 40 \\ + 200 \\ \hline 264.12 \end{array} $	$ \begin{array}{r} 12.4 \\ \times 21.3 \\ \hline 3.72 \\ 12.40 \\ + 248.00 \\ \hline 264.12 \end{array} $

Discuss each method with your partner while answering the following questions.

- In Sergio's work, which parts of the factors were multiplied to get each partial product indicated in the table below?

Partial Product	Parts of Factors Multiplied
0.12	___ $\times$ ___
0.6	___ $\times$ ___
3	___ $\times$ ___

- b. To which parts of the area model do these partial products relate?



- c. Where do the values of 3.72, 12.4, and 248 come from in Alicia's work that shows the standard algorithm?

- _____ is the product that results from multiplying by each part of 12.4.

- ☐ 0.3
- ☐ 1
- ☐ 20

- _____ is the product that results from multiplying by each part of 12.4.

- ☐ 0.3
- ☐ 1
- ☐ 20

- _____ is the product that results from multiplying by each part of 12.4.

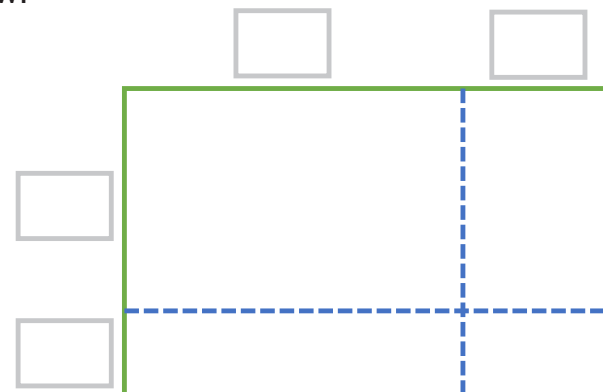
- ☐ 0.3
- ☐ 1
- ☐ 20

- d. Draw lines to indicate connections between the partial products found in each of Sergio and Alicia's methods.
- e. In each method, why are the numbers between the horizontal lines aligned vertically?



1.4.4 Guided Practice: Multiplying Decimals to the Thousandths

1. Find the product $(32.5)(6.02)$ using the area model below.



2. With your partner, determine who will be Partner A and Partner B.

- a. Find the product of 3.621×9.7 using the method indicated. Show your work in the space provided.

Partner A Partial Products	Partner B Standard Algorithm

- b. Compare your work with your partner to verify that you both found the same product. Discuss any errors and make corrections as needed. Once you have come to an agreement, write your partner's work in your workbook in the space provided.
- c. Discuss with your partner the strengths and weaknesses of using each method.
- d. Which method do you prefer for multiplying decimals to the thousandths?



1.4.5 Try It: Multiplying Decimals to the Thousandths

1. Find each product using your preferred method. Show your work.
 - a. $25.36 \cdot 14.8$
 - b. 19.008×3.2
 - c. $(36.3)(0.74)$



1.4.6 Your Turn!

1. Find each product.

a. 105.46×3.8

b. $8.7 \cdot 0.475$

c. $(7.7)(102.65)$

d. 1.561×12.8



1.4.7 Wrap-Up: Ticket Out the Door

1. Find each product.

a. $(6.36)(12.27)$

b. $8.12 \cdot 29.4$